Dr. Mohd Hamza Naim Shaikh

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34 peer-reviewed publications (25 first-authored) | Google Scholar: 300+ citations, h-index 10

Professional Summary

Research Fellow at Queen's University Belfast working on the physical layer of beyond-5G/6G wireless networks, with research and teaching experience across India, Kazakhstan and the United Kingdom, including appointments at University College London and Nazarbayev University. Research spans reconfigurable intelligent surfaces (RIS and STAR-RIS), fluid antenna systems and fluid RIS, non-orthogonal and rate-splitting multiple access, integrated sensing and communication (ISAC), and hardware-impaired transceivers, combining communication theory, stochastic analysis, optimization and learning-based methods. Author of 34 peer-reviewed publications, including first-author articles in IEEE Transactions on Wireless Communications, IEEE Transactions on Vehicular Technology, IEEE Wireless Communications Letters and IEEE Systems Journal, and recipient of the Best PhD Dissertation Award (IIIT-Delhi, 2023).

Research Highlights

- 34 peer-reviewed publications: 14 journal articles, 18 conference papers and 2 book chapters; 25 as first author.
- First-author articles in IEEE TWC, IEEE TVT, IEEE WCL (2), IEEE Systems Journal (2), IEEE OJ-COMS, IEEE OJSP and IEEE Access.
- Google Scholar: 300+ citations, h-index 10.
- Best PhD Dissertation Award, IIIT-Delhi (2023); Visvesvaraya PhD Fellowship, MeitY, Government of India (2017–2022).
- International research experience in India, Kazakhstan and the United Kingdom, with collaborators at UCL, KAUST, KFUPM, the University of Southampton, Nazarbayev University and IIIT-Delhi.
- Reviewer for IEEE TWC, TVT, IoTJ, WCL and Communications Letters; TPC member for IEEE ICC, VTC, ANTS, FNWF and WF-IoT.

Academic Appointments

Research Fellow 2026 – Present

Queen's University Belfast (QUB), United Kingdom

Research on RIS-enabled and fluid antenna system-assisted next-generation wireless networks, with emphasis on analytical modelling, optimization and performance evaluation.

Research Fellow 2025 – 2026

University College London (UCL), United Kingdom

Fluid reconfigurable intelligent surfaces (FRIS) for NOMA, and port selection for compact ultra-massive antenna arrays (fluid-antenna multiple access); results published in IEEE Wireless Communications Letters. Session Chair, WI-NOW 2025 International Workshop.

Postdoctoral Research Assistant 2022 – 2025

Nazarbayev University, Kazakhstan

RIS- and STAR-RIS-assisted NOMA, UAV-enabled wireless systems, integrated sensing and communication (ISAC) and mmWave communications; first-author work published in IEEE TWC and IEEE TVT. Co-supervised undergraduate researchers, resulting in Q1 journal publications.

Teaching Assistant 2016 – 2020

IIIT-Delhi, India

Teaching and laboratory support for communication engineering courses. Recipient of the Teaching Assistant Award for Mobile Communication.

Education

PhD in Communication Engineering 2016 – 2022

IIIT-Delhi, India

Thesis: *Performance Analysis of Future Wireless Communication Systems under Residual Hardware Impairments*
Advisors: Prof. Vivek Ashok Bohara and Prof. Anand Srivastava
Best PhD Dissertation Award (2023); Visvesvaraya PhD Fellow

M.Tech in Communication and Information Systems 2014 – 2016

Aligarh Muslim University, India

B.Tech in Electronics Engineering 2010 – 2014

Aligarh Muslim University, India

Research Interests

- Reconfigurable Intelligent Surfaces (RIS, STAR-RIS)
- Fluid and Movable Antenna Systems
- Integrated Sensing and Communication (ISAC)
- NOMA and Rate-Splitting Multiple Access
- Physical Layer Security
- Hardware-Impaired Transceivers
- UAV and Vehicular Communications
- Massive MIMO and Full-Duplex Systems
- Signal Processing and Optimization
- Learning-Based Wireless Networks for 6G

Teaching

Teaching Assistant at IIIT-Delhi (2016–2020) for communication engineering courses; Teaching Assistant Award for Mobile Communication (2018–2019); Institute Teaching Fellowship (2016–2017).

Teaching Areas

- Wireless Communication
- Digital Communication
- Mobile Communication
- Communication Theory
- Signals and Systems
- Probability and Random Processes
- Digital Signal Processing
- Engineering Mathematics
- Machine Learning
- 5G / 6G Wireless Networks

Supervision and Mentoring

- Co-supervised undergraduate researchers at Nazarbayev University (2022–2025), resulting in Q1 journal publications.

Awards and Fellowships

- Best PhD Dissertation Award, IIIT-Delhi 2023
- Visvesvaraya PhD Fellowship, MeitY, Government of India 2017–2022
- Teaching Assistant Award, Mobile Communication, IIIT-Delhi 2018–2019
- Institute Teaching Fellowship, IIIT-Delhi 2016–2017
- GATE Scholarship, AICTE, Government of India 2014–2016

National-Level Qualifications

- UGC-NET Qualified (Electronic Science) 2017
- GATE Qualified (Electronics and Communication Engineering) 2014 and 2015

Academic Service and Professional Activities

Journal Reviewer

- IEEE Transactions on Wireless Communications
- IEEE Transactions on Vehicular Technology
- IEEE Internet of Things Journal
- IEEE Systems Journal
- IEEE Communications Letters
- IEEE Wireless Communications Letters
- IEEE Access
- Vehicular Communications
- Wireless Networks

Technical Program Committee Member

- IEEE International Conference on Communications (ICC) 2025
- IEEE ANTS 2020 and 2023–2026
- IEEE Vehicular Technology Conference (VTC)
- IEEE Future Networks World Forum (FNWF)
- IEEE World Forum on Internet of Things (WF-IoT)
- ICCSPA 2026

Session Chair

- WI-NOW 2025 International Workshop, University College London
- IEEE ANTS 2020

Professional Memberships

Member, IEEE | Member, IEEE Communications Society (ComSoc) | Member, IEEE Vehicular Technology Society (VTS)

Research Collaborations

- University College London, United Kingdom
- University of Southampton, United Kingdom
- Nazarbayev University, Kazakhstan
- KAUST, Saudi Arabia
- KFUPM, Saudi Arabia
- IIIT-Delhi, India

Publications

34 peer-reviewed publications (14 journal articles, 18 conference papers, 2 book chapters), 25 of them first-authored. Numbering runs from oldest to newest within each category; titles link to the publisher's page.

Journal Articles

[J14] **M. H. N. Shaikh**, K.-K. Wong, F. R. Ghadi, C.-B. Chae, and Y. Zhang, "Sum-Rate Maximization for FRIS-Enabled NOMA Communications," *IEEE Wireless Communications Letters*, 2026.

[J13] **M. H. N. Shaikh**, K.-K. Wong, F. R. Ghadi, *et al.*, "On Performance of FRIS-Enabled Uplink NOMA," *IEEE Wireless Communications Letters*, 2026.

[J12] T. Altay, **M. H. N. Shaikh**, A. Çelik, A. M. Eltawil, and G. Nauryzbayev, "Aerial RIS-Aided Uplink NOMA: A DRL-Aided Resource Allocation Framework," *IEEE Transactions on Cognitive Communications and Networking*, 2026.

[J11] **M. H. N. Shaikh**, K. M. Rabie, T. Y. Elganimi, X. Li, M. S. Hashmi, and G. Nauryzbayev, "Multi-STAR-RIS NOMA: Clustering, RIS Assignment and Power Optimization," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 4, pp. 6389–6405, 2025.

[J10] A. Serikov, **M. H. N. Shaikh**, and G. Nauryzbayev, "Enhancing Vehicular NOMA Communication Security through Reconfigurable Intelligent Surfaces," *Vehicular Communications*, vol. 50, Art. no. 100841, 2024.

[J9] T. Toregozhin, **M. H. N. Shaikh**, A. Almagambetov, and G. Nauryzbayev, "Performance of STAR-RIS-Aided Cooperative NOMA Networks under Nakagami-m Fading," *Ad Hoc Networks*, vol. 156, Art. no. 103399, 2024.

[J8] **M. H. N. Shaikh**, A. Çelik, A. M. Eltawil, and G. Nauryzbayev, "Grant-Free NOMA Through Optimal Partitioning and Cluster Assignment in STAR-RIS Networks," *IEEE Transactions on Wireless Communications*, vol. 23, no. 8, pp. 10166–10181, Aug. 2024.

[J7] **M. H. N. Shaikh**, V. A. Bohara, A. Srivastava, and G. Ghatak, "An Energy Efficient Dual IRS-Aided Outdoor-to-Indoor Communication System," *IEEE Systems Journal*, vol. 17, no. 3, pp. 3718–3729, 2023.

[J6] **M. H. N. Shaikh**, V. A. Bohara, and A. Srivastava, "On the Performance of Dual IRS-Aided Communication System under Non-Ideal Transceiver over Nakagami-m Fading Channels," *Internet Technology Letters*, vol. 6, no. 3, 2023.

[J5] K. Joshi, **M. H. N. Shaikh**, S. A. Naqvi, and V. A. Bohara, "On the Performance of IRS-Assisted OFDM System with Non-Ideal Oscillator and Amplifier," *Internet Technology Letters*, vol. 5, no. 4, 2022, doi: 10.1002/itl2.380.

[J4] **M. H. N. Shaikh**, V. A. Bohara, A. Srivastava, and G. Ghatak, "A Downlink RIS-Aided NOMA System With Hardware Impairments: Performance Characterization and Analysis," *IEEE Open Journal of Signal Processing*, vol. 3, 2022.

[J3] **M. H. N. Shaikh**, V. A. Bohara, A. Srivastava, and G. Ghatak, "Performance Analysis of Intelligent Reflecting Surface-Assisted Wireless System With Non-Ideal Transceiver," *IEEE Open Journal of the Communications Society*, vol. 2, pp. 671–686, 2021.

[J2] **M. H. N. Shaikh**, V. A. Bohara, and A. Srivastava, "Performance Analysis of a Full-Duplex MIMO Decode-and-Forward Relay System With Self-Energy Recycling," *IEEE Access*, vol. 8, pp. 226248–226266, 2020.

[J1] **M. H. N. Shaikh**, V. A. Bohara, P. Aggarwal, and A. Srivastava, "Energy Efficiency Evaluation for Downlink Full-Duplex Nonlinear MU-MIMO-OFDM System With Self-Energy Recycling," *IEEE Systems Journal*, vol. 14, no. 3, pp. 3313–3324, 2020.

Conference Papers

[C18] **M. H. N. Shaikh**, A. Çelik, S. Arzykulov, *et al.*, "Optimizing Deployment and Partitioning Strategies for Aerial RIS-Aided Uplink NOMA under Residual Hardware Impairments," Conference paper,

2025.

[C17] **M. H. N. Shaikh**, A. Çelik, A. M. Eltawil, and G. Nauryzbayev, "Sense and Jam: ISAC-Aided Robust Physical Layer Security," in *Proc. 58th Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, USA, Oct. 2024, pp. 653–657.

[C16] **M. H. N. Shaikh**, A. Çelik, A. M. Eltawil, and G. Nauryzbayev, "On the Optimization of Virtual RIS Partitioning for Grant-Free Non-Orthogonal Multiple Access," in *Proc. IEEE Global Communications Conference (GLOBECOM)*, Kuala Lumpur, Malaysia, Dec. 2023, pp. 7219–7224.

[C15] T. Toregozhin, **M. H. N. Shaikh**, and G. Nauryzbayev, "Performance Analysis and Enhancement through Cooperative NOMA in STAR-RIS-Aided Systems," in *Proc. International Balkan Conference on Communications and Networking (BalkanCom)*, 2023.

[C14] **M. H. N. Shaikh**, V. A. Bohara, and A. Srivastava, "IRS-Aided Communication System with Phase Noise and Hardware Impairments: Performance Analysis and Characterization," in *Proc. International Conference on COMmunication Systems & NETworkS (COMSNETS)*, Bengaluru, India, Jan. 2023, pp. 392–396.

[C13] K. Lata, V. A. Bohara, A. Srivastava, and **M. H. N. Shaikh**, "Intelligent Reflecting Surfaces Versus Optical Mirrors: Performance Comparison for Indoor Physical Layer Security Environments," in *Proc. IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, Dec. 2022, pp. 314–319.

[C12] **M. H. N. Shaikh**, S. Arzykulov, A. Çelik, A. M. Eltawil, and G. Nauryzbayev, "Performance of RIS-Empowered NOMA-Based D2D Communication under Nakagami-m Fading," in *Proc. IEEE 96th Vehicular Technology Conference (VTC2022-Fall)*, London, UK, Sep. 2022, pp. 1–5.

[C11] **M. H. N. Shaikh**, K. M. Rabie, X. Li, T. A. Tsiftsis, and G. Nauryzbayev, "On the Performance of Dual RIS-Assisted V2I Communication under Nakagami-m Fading," in *Proc. IEEE 96th Vehicular Technology Conference (VTC2022-Fall)*, London, UK, Sep. 2022, pp. 1–5.

[C10] **M. H. N. Shaikh**, V. A. Bohara, A. Srivastava, and G. Ghatak, "On the Performance of RIS-Aided NOMA System with Non-Ideal Transceiver over Nakagami-m Fading," in *Proc. IEEE Wireless Communications and Networking Conference (WCNC)*, Austin, TX, USA, Apr. 2022, pp. 1737–1742.

[C9] **M. H. N. Shaikh**, V. A. Bohara, A. Srivastava, and G. Ghatak, "Intelligent Reflecting Surfaces Versus Full-Duplex Relaying: Performance Comparison for Non-Ideal Transmitter Case," in *Proc. IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Sep. 2021, pp. 513–518.

[C8] **M. H. N. Shaikh**, V. A. Bohara, and A. Srivastava, "Performance Enhancement in Full-Duplex AF Relay System through Smart Antenna Allocation," in *Proc. IEEE 3rd 5G World Forum (5GWF)*, Sep. 2020, pp. 303–308.

[C7] **M. H. N. Shaikh**, V. A. Bohara, A. Srivastava, and G. Ghatak, "EE Enhancement in FD MIMO Relay System through Adaptive Antenna Allocation and Self-Energy Recycling," in *Proc. IEEE International Conference on Communications Workshops (ICC Workshops)*, Jun. 2020, pp. 1–7.

[C6] **M. H. N. Shaikh**, V. A. Bohara, and A. Srivastava, "Spectral Analysis of a Nonlinear WOLA-OFDM System with DPD," in *Proc. National Conference on Communications (NCC)*, Feb. 2020, pp. 1–4, doi: 10.1109/NCC48643.2020.9056066.

[C5] **M. H. N. Shaikh**, V. A. Bohara, and A. Srivastava, "Energy Efficiency Enhancement in Full-Duplex Relay System through Adaptive Antenna Allocation," in *Proc. IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, Dec. 2019, pp. 1–6.

[C4] **M. H. N. Shaikh**, V. A. Bohara, P. Aggarwal, and A. Srivastava, "On EE-SE Trade-Off for Downlink Full Duplex MISO Systems with Self-Energy Recycling," in *Proc. IEEE 89th Vehicular Technology Conference (VTC2019-Spring)*, Apr. 2019, pp. 1–5.

[C3] **M. H. N. Shaikh**, A. Agarwal, P. Aggarwal, and V. A. Bohara, "On the Spectral Content of Nonlinear Carrier Aggregated Windowed OFDM Systems," in *Proc. IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, Dec. 2018, pp. 1–5.

[C2] G. Chandel, O. Farooq, **M. H. N. Shaikh**, and M. Shanir P. P., "Seizure Detection in Neonatal EEG Signals Using EMD Based Features," Conference paper, 2017.

[C1] A. Mohammad, F. Azeem, M. Noman, and **M. H. N. Shaikh**, "A Simple Approach of ECG Beat Classification," Conference paper, 2016.

Book Chapters

[B2] S. Rajak, V. C. Dodda, Lakshmi, *et al.*, "Evolution of Intelligent Transportation Systems for the Next Generation: IRS Enabled 6G," Book chapter, 2026.

[B1] **M. H. N. Shaikh**, O. Farooq, and G. Chandel, "EMD Analysis of EEG Signals for Seizure Detection," in *Advances in System Optimization and Control, Lecture Notes in Electrical Engineering*, Springer, 2018, pp. 189–196, doi: 10.1007/978-981-13-0665-5_16.

Preprints and Manuscripts Under Review

[P3] M. Asif, A. Ihsan, I. Muhammad, *et al.*, "Exploiting Movable-Element STARS for Rate Splitting Multiple Access," Preprint, 2026.

[P2] M. Asif, A. Ihsan, I. Muhammad, *et al.*, "Robust Transmission Design for RIS-Assisted RSMA-SWIPT Systems With Movable Antennas Under Hardware Distortions," Preprint, 2026.

[P1] C. Rao, K.-K. Wong, **M. H. N. Shaikh**, H. Hong, H. Shin, and Y. Zhang, "Geometric Port Selection in CUMA Systems," arXiv preprint arXiv:2509.20299, 2025.

Research Vision

To develop intelligent, sustainable and socially impactful communication technologies by integrating communication, sensing, networking and artificial intelligence for future digital societies.

"Building Intelligent and Sustainable Digital Futures through ICT"

References

Available upon request.